GoPhish & MailHog

Local Security Awareness Simulation

Environment: Kali Linux running in Oracle VirtualBox
Simulation: LAB-CAMPAIGN-01
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Scope and Safety Statement

This project involved a controlled, localhost-only security awareness simulation conducted in a Kali Linux virtual machine using GoPhish and MailHog. The primary objective was to demonstrate the technical workflow of a phishing awareness exercise, including email template configuration, local email delivery, tracking-link functionality, landing-page hosting, and campaign event monitoring.

The simulation used student1@example.test as a fictional test recipient to prevent the exercise from involving real individuals or external email accounts. MailHog was configured as a local SMTP testing service to capture and display outgoing test messages within the lab environment rather than deliver them to external recipients. GoPhish was configured to use localhost addresses for its administrative interface and phishing simulation server, keeping the services confined to the local machine.

The campaign email contained a link to a locally hosted security awareness landing page. This page provided educational guidance on identifying suspicious emails, checking sender addresses, inspecting links before opening them, recognizing unexpected or urgent requests, and reporting potentially malicious messages through appropriate channels. The landing page did not contain a credential submission form, request passwords, or collect sensitive authentication information.

During testing, GoPhish recorded one email sent, one email opened, and one link clicked. These events were generated through the author's own testing activities to verify that the campaign tracking mechanism and landing page functioned as intended. Therefore, the recorded results demonstrate successful technical validation of the lab setup rather than evidence of a real user being deceived by a phishing attempt.

No submitted-data events were recorded, and no credentials were collected during the exercise. The project was limited to testing within the local environment and was intended solely for educational purposes, technical practice, and the development of cybersecurity awareness skills. These safeguards helped ensure that the simulation demonstrated phishing-related techniques without exposing external users to the test or collecting their sensitive information.

Executive Summary

This project involved the configuration and testing of a local phishing-awareness simulation environment using GoPhish and MailHog within a Kali Linux virtual machine running on Oracle VirtualBox. The primary objective was to demonstrate how phishing simulation tools can be used in a controlled cybersecurity laboratory to examine email delivery, simulate user interaction with a tracking link, host an educational landing page, and monitor campaign events. The activity also provided practical experience in configuring cybersecurity tools, troubleshooting technical issues, and documenting the results of a security testing exercise.

GoPhish was configured to manage the campaign, generate the simulated awareness email, provide a tracking link, host the landing page, and record campaign events. MailHog served as the local SMTP testing service, capturing outgoing test messages for inspection without delivering them to an external mailbox. Both services were configured to use localhost addresses, keeping the exercise within the local laboratory environment. A fictional test recipient, student1@example.test, was used to prevent the campaign from involving real recipients.

The simulation successfully demonstrated the essential components of the testing workflow. The campaign email was captured in MailHog, the tracking URL directed the browser to the locally hosted Security Awareness Exercise landing page, and GoPhish recorded the resulting campaign activity. The landing page provided educational guidance on identifying suspicious emails, examining sender addresses, inspecting links, recognizing unexpected requests, and reporting potentially malicious messages. It did not request or collect passwords or other credentials.

The GoPhish results dashboard recorded **one email sent, one email opened, and one link clicked**. It also recorded **zero submitted-data events and zero reported-email events**. These results confirmed that the principal technical components of the local simulation were functioning as intended. However, the open and click events resulted from the author's own testing activities and should not be interpreted as evidence of actual user susceptibility to phishing.

Overall, the project demonstrated the practical use of GoPhish and MailHog for conducting a controlled security-awareness exercise. It strengthened the author's understanding of local SMTP testing, email-template configuration, landing-page deployment, tracking mechanisms, campaign monitoring, and evidence collection. The exercise also emphasized the importance of maintaining a clearly defined testing scope, restricting services to the local environment, avoiding the collection of sensitive information, and interpreting campaign metrics accurately. The completed setup provides a foundation for further authorized cybersecurity laboratory exercises and the development of phishing-awareness and defensive security skills.

Objectives and Scope

2.1 Project Objectives

The primary objective of this project was to establish and validate a controlled phishing-awareness simulation environment using GoPhish and MailHog within a Kali Linux virtual machine. The exercise was designed to provide practical experience with email-based security simulations, local service configuration, campaign monitoring, and the documentation of cybersecurity testing results.

The specific objectives of the project were as follows:

1. **Configure GoPhish for localhost-only operation.** Configure the GoPhish administrative interface and phishing simulation server to listen on local addresses. This configuration was intended to restrict access to the laboratory environment and reduce the risk of unintentionally exposing the services to other systems or external networks.

2. **Deploy MailHog as a local SMTP testing service.** Start MailHog and configure it to receive test messages through a local SMTP connection. Inspect captured messages through its web interface to verify that the email-generation and delivery workflow operated correctly without requiring delivery to an external mailbox.

3. **Send and inspect a test email using a fictional recipient.** Use student1@example.test as the test recipient to validate the email-sending process in a controlled environment. Examine the captured message to verify its subject, sender information, body content, and tracking link.

4. **Verify the campaign tracking URL and awareness landing page.** Confirm that the URL generated by GoPhish directed the browser to the locally hosted security awareness landing page. Verify that the page loaded successfully and presented educational information about recognizing suspicious emails without requesting or collecting credentials.

5. **Review and document campaign metrics.** Examine the GoPhish campaign results dashboard to identify recorded events, including emails sent, emails opened, links clicked, submitted data, and reported emails. Document the results and interpret them in the context of the controlled test rather than treating them as evidence of real-world user behavior.

6. **Develop practical cybersecurity and documentation skills.** Gain hands-on experience with local service configuration, SMTP testing, email-template management, landing-page deployment, campaign event analysis, troubleshooting, screenshot collection, and technical report preparation.

2.2 Project Scope

The project covered the setup, configuration, and functional testing of GoPhish and MailHog within a local Kali Linux virtual machine. The work included configuring local service addresses, creating a simulated awareness email, capturing the email through MailHog, hosting an educational landing page, following the campaign tracking URL, and reviewing the resulting campaign metrics.
The exercise used a fictional test recipient and was intended for educational and technical validation purposes. The landing page focused on security awareness and did not include a credential submission form. The collected evidence consisted of application screenshots, the captured test email, the displayed landing page, and the GoPhish campaign results.

The project was limited to demonstrating the technical workflow of a controlled phishing-awareness simulation. It was not intended to assess the behavior of actual users, measure an organization's susceptibility to phishing, or evaluate the effectiveness of enterprise email security controls.

2.3 Out of Scope

To maintain a clearly defined and controlled testing environment, the following activities were excluded from the project:

- **External email recipients:** The campaign was not intended to target real individuals or send simulated phishing messages to external email accounts.
- **Public exposure of the phishing server:** The GoPhish phishing server was configured for localhost-only operation and was not intended to be made publicly accessible.
- **Credential harvesting:** The exercise did not involve collecting usernames, passwords, authentication tokens, or other sensitive authentication information.
- **Password collection or submission:** The awareness landing page did not request passwords or provide a credential submission mechanism.
- **Unauthorized targeting:** The project did not include testing against individuals, organizations, or systems outside the defined laboratory environment.
- **Claims of real-user deception:** The recorded open and click events were generated during the author's own testing. They were not treated as evidence that a real user had been deceived or had failed to recognize a phishing attempt.
- **Enterprise-level effectiveness testing:** The exercise did not evaluate production email infrastructure, organizational reporting procedures, or the effectiveness of deployed security controls.

2.4 Scope and Safety Considerations

The project maintained a local testing boundary by using localhost addresses for the GoPhish services and MailHog. The fictional recipient address allowed the email workflow to be tested without involving a real recipient, while MailHog provided a means of inspecting messages without external email delivery.

These restrictions helped keep the activity focused on technical learning, controlled simulation, and security awareness. They also ensured that the reported results remained aligned with the evidence collected during the exercise.

The project was considered technically successful when the test email was captured by MailHog, the campaign tracking URL opened the local awareness page, and GoPhish recorded the expected campaign events without collecting credentials.

Lab Architecture and Configuration

3.1 Overview of the Lab Environment

The laboratory environment was implemented using Kali Linux running in an Oracle VirtualBox virtual machine. GoPhish served as the phishing simulation and campaign management platform, while MailHog functioned as the local SMTP testing service used to capture and inspect outgoing test emails.

The two applications served different purposes within the workflow. GoPhish generated the campaign email, managed the sending profile, hosted the security awareness landing page, and recorded campaign events. MailHog received the outgoing test message through a local SMTP connection and displayed the captured email through its web interface.

The services were configured to use the loopback address, 127.0.0.1, which refers to the local machine. This allowed the components to communicate within the Kali Linux environment without requiring the test email to be delivered to an external mailbox.

3.2 Components and Configuration

Component	Local address	Purpose
GoPhish administrative interface	https://127.0.0.1:3333	Provides access to campaign management, email templates, sending profiles, landing pages, and campaign results.
GoPhish phishing server	http://127.0.0.1:8080	Hosts the local awareness landing page and provides the campaign tracking URL used during testing.
MailHog SMTP service	127.0.0.1:1025	Receives outgoing test email from GoPhish for local inspection instead of delivering it to an external mail server.
MailHog web interface	http://127.0.0.1:8025	Displays captured email messages and allows inspection of their headers, content, and links.

3.3 GoPhish Configuration

GoPhish was configured with two primary services: the administrative interface and the phishing simulation server.

The administrative interface was bound to 127.0.0.1:3333 and used HTTPS to provide access to the GoPhish management dashboard. Through this interface, the campaign email template, sending profile, landing page, campaign configuration, and event results could be managed.

The phishing server was configured to listen on 127.0.0.1:8080 using HTTP for the local simulation. This server hosted the security awareness landing page and handled the tracking URL generated for the campaign recipient. The landing page provided educational guidance on recognizing suspicious emails and did not request or collect credentials.

Binding both GoPhish services to the loopback address restricted their intended access to the local machine. This configuration was appropriate for the project because the exercise did not require access from other devices or external networks.

3.4 MailHog Configuration

MailHog was configured as the laboratory's local SMTP sink. Its SMTP service listened on 127.0.0.1:1025, allowing GoPhish to submit test messages through a local connection.

The MailHog web interface was accessible at http://127.0.0.1:8025. It provided a convenient way to verify that messages had been received and to inspect the email subject, sender, recipient, body content, and tracking URL.

The fictional recipient address student1@example.test was used for the controlled test. MailHog captured the test message locally, allowing the email workflow to be examined without relying on an external email provider.

3.5 Communication Workflow

The laboratory workflow consisted of the following stages:

1. **Campaign preparation:** The email template, sending profile, awareness landing page, and fictional test recipient were configured in GoPhish.
2. **Email submission:** GoPhish submitted the campaign message to MailHog through the local SMTP endpoint at 127.0.0.1:1025.
3. **Message inspection:** MailHog displayed the captured message through its web interface at 127.0.0.1:8025.
4. **Tracking-link verification:** The tracking URL in the message was opened in the browser, directing the test session to the local GoPhish phishing server.
5. **Landing-page display:** The awareness page loaded at http://127.0.0.1:8080 and displayed guidance about identifying suspicious emails.
6. **Event recording:** GoPhish recorded the campaign activity, which was subsequently reviewed through the administrative dashboard.

3.6 Security and Configuration Considerations

The use of loopback addresses helped isolate the exercise from other machines on the network. MailHog also provided a safe method of examining outgoing messages without sending them to external recipients. The fictional test recipient and the absence of a credential submission form further limited the scope of the exercise.

The administrative interface used HTTPS with a locally configured certificate. Because the certificate was self-signed, the browser could display a certificate warning when the interface was accessed. Such a warning is expected in many local test environments, but it should not be interpreted as proof that the service is secure. The certificate's trust status and the service's actual configuration must be considered separately.

The phishing simulation server used HTTP for this local exercise. Consequently, the setup was intended for controlled testing rather than deployment in a production environment. Any future expansion beyond localhost would require a separate security review, explicit authorization, appropriate network restrictions, and secure transport configuration.

Overall, the laboratory architecture provided a simple and controlled environment for demonstrating email simulation, local SMTP capture, landing-page hosting, and campaign event monitoring without involving external recipients or collecting sensitive authentication information.

Methodology

4.1 Overview

The activity followed a controlled, step-by-step procedure to configure and test a local phishing-awareness simulation using GoPhish and MailHog in a Kali Linux environment. The methodology focused on verifying the functionality of the email delivery system, testing the campaign tracking mechanism, and confirming that the awareness landing page displayed the intended educational content.

All tests were conducted within the local laboratory environment using a fictional recipient address. The campaign was designed for security-awareness testing and did not require or collect passwords, account credentials, or other sensitive information.

4.2 Procedure

Step 1: Start GoPhish and Verify Its Services

GoPhish was launched from its project directory in Kali Linux. The startup output was reviewed to confirm that the administrative interface and phishing server started successfully.

The administrative interface was configured to listen on 127.0.0.1:3333, while the phishing server listened on 127.0.0.1:8080. These local addresses allowed the services to be accessed from the same machine.

Verification: The startup logs were checked for messages indicating that both listeners had started. The GoPhish administrative interface was then accessed through a local web browser.

Purpose: To ensure that the campaign management interface and the local campaign server were operational before proceeding.

Step 2: Start MailHog and Verify the Email Testing Environment

MailHog was started with its SMTP service bound to 127.0.0.1:1025 and its web interface bound to 127.0.0.1:8025.

The SMTP service served as the destination for outgoing test emails generated by GoPhish. The web interface provided a convenient way to inspect received messages without sending them through an external email service.

Verification: MailHog was accessed through http://127.0.0.1:8025 to confirm that its web interface was available.

Purpose: To establish a local email-testing environment where messages could be inspected safely.

Step 3: Perform a Direct SMTP Verification Test

A direct SMTP verification message was sent to the fictional recipient address student1@example.test. The message was intended to confirm that MailHog could receive and display email submitted through its SMTP service.

After the message was sent, the MailHog interface was checked for the test email. The displayed message details were reviewed to verify successful receipt.

Verification: The test message appeared in the MailHog inbox and could be opened for inspection.

Purpose: To verify the local SMTP delivery path independently before configuring and launching the GoPhish campaign.

Step 4: Configure the Email Template and Sending Profile

The GoPhish email template named LAB-Email-01 was created for the awareness exercise. Its subject was configured as Security Awareness Exercise: Account Notice, with the sender address set to lab@example.test.

The template included the GoPhish tracking placeholder {{.URL}}, which was used to generate the campaign-specific link. The HTML and plain-text versions of the email were treated as separate content formats to ensure that the message would display correctly in different email views.

A sending profile named LAB-MailHog-Local was configured to use the local SMTP endpoint at 127.0.0.1:1025. Authentication was not required for this isolated local configuration.

Verification: The template and sending profile settings were reviewed before campaign launch. The resulting message was subsequently inspected in MailHog.

Purpose: To prepare the email content and configure GoPhish to deliver campaign messages to the local testing service.

Step 5: Configure the Security-Awareness Landing Page

A landing page named LAB-Awareness-01 was configured to display security-awareness guidance. The page was intended to explain the importance of recognizing suspicious messages, checking links before opening them, and reporting potentially malicious emails.

The page did not include a credential submission form. Its purpose was educational rather than to collect user information.

Verification: The landing page was opened in a browser to confirm that the expected awareness content loaded and was readable.

Purpose: To provide an educational destination for the campaign link without requesting or collecting sensitive information.

Step 6: Create and Launch the Test Campaign

A GoPhish campaign named LAB-CAMPAIGN-01 was configured using the LAB-Email-01 template, the LAB-MailHog-Local sending profile, the LAB-Awareness-01 landing page, and the fictional recipient student1@example.test.

The campaign was configured to use http://127.0.0.1:8080 as its local campaign URL. After the configuration was reviewed, the campaign was launched within the laboratory environment.

Verification: The campaign appeared in the GoPhish campaign list, and the message was checked in MailHog to confirm that the email had been received.

Purpose: To test the integration between GoPhish, MailHog, and the awareness landing page using a controlled test recipient.

Step 7: Test the Campaign Tracking Link

The campaign email was inspected in MailHog, and the generated campaign link was opened in the local browser. The browser was used to verify that the link directed to the configured awareness landing page.

The page was checked for successful loading and the expected educational content. The test also allowed GoPhish to record campaign interactions associated with the tracked link.

Verification: The awareness landing page loaded successfully, and the campaign results dashboard recorded the corresponding interaction events.

Purpose: To verify that the generated link, landing page, and tracking mechanism worked together as expected.

Step 8: Review Campaign Results and Preserve Evidence

After the test interactions were completed, the GoPhish campaign results dashboard was reviewed. The displayed email delivery, opening, link-click, data-submission, and reporting counts were recorded.

Screenshots of the relevant configuration pages, MailHog messages, landing page, and campaign results were collected to document the setup and outcome of the exercise.

Verification: The dashboard showed one email sent, one email opened, one link clicked, zero submitted data, and zero reported emails.

The opening and link-click events were generated during the test activity. They should therefore be interpreted as results of the controlled simulation, not as evidence that an independent user interacted with a phishing email.

Purpose: To document the observed results, preserve supporting evidence, and provide a basis for the findings and recommendations presented in the following sections.

4.3 Evidence Collection

Screenshots were used to support the methodology and demonstrate the successful completion of the principal test stages. The evidence included:

- GoPhish startup and configuration information.
- MailHog's SMTP verification message.
- The configured email template and sending profile.
- The security-awareness landing page.
- The campaign listing and results dashboard.
- The recorded campaign interaction counts.

The screenshots serve as supporting evidence of the observed configuration and test outcomes. They do not, by themselves, establish the security of a production email system or demonstrate how an actual user would respond to a phishing attempt.

4.4 Ethical and Safety Considerations

The experiment was conducted as a local cybersecurity laboratory exercise. A fictional recipient was used, and the test email was directed to MailHog rather than an external email recipient. The landing page provided awareness information without requesting credentials.

The procedure was limited to the intended test environment. No claim is made that the results represent a real-world phishing campaign, an external email-delivery test, or a comprehensive security assessment.

These controls helped keep the exercise focused on learning how phishing-simulation tools, local SMTP testing, campaign tracking, and security-awareness content operate together.

Results

5.1 Campaign Performance Summary

The GoPhish campaign LAB-CAMPAIGN-01 was executed in the local laboratory environment using MailHog as the SMTP testing service. The campaign results dashboard recorded the following metrics after the author's controlled test activity.

Metric	Observed Result	Interpretation
Emails sent	1	GoPhish recorded one email as sent to the configured test recipient.
Emails opened	1	One email-open event was recorded during testing.
Links clicked	1	One tracked-link click event was recorded.
Submitted data	0	No data-submission events were recorded.
Emails reported	0	No email-reporting events were recorded.

5.2 Analysis of Observed Results

Email Delivery

The campaign dashboard recorded one email sent. The message was also inspected in MailHog, confirming that the local email-testing workflow was functioning. This demonstrated that GoPhish could submit a campaign email to the configured local SMTP service.

Because MailHog captures messages locally, this result demonstrates successful delivery to the test inbox rather than delivery to an external mailbox or a real email recipient.

Email Open Tracking

The dashboard recorded one email-open event. This indicates that GoPhish registered an opening-related tracking event during the controlled test.

Email-open tracking commonly relies on a tracking element embedded in the message. Such events are not definitive proof that a human read the email because email clients, image-loading behavior, and automated systems can affect tracking results. In this exercise, the event was associated with the author's own test activity.

Link-Click Tracking

One link-click event was recorded after the campaign URL was opened in the local browser. The configured security-awareness landing page loaded successfully and displayed its educational content.

This result supports the conclusion that the generated campaign link and local landing page were connected correctly and that GoPhish recorded the test interaction. It does not establish that an independent user would click the same link in a real phishing scenario.

Data Submission

The dashboard recorded zero submitted-data events. This was consistent with the design of LAB-Awareness-01, which presented security-awareness guidance without a credential-submission form.

The absence of data submissions was an intended safety outcome. The exercise was designed to demonstrate campaign tracking and awareness-page functionality without collecting passwords or other sensitive information.

Email Reporting

The dashboard recorded zero reported emails. No reporting event was recorded during the test. Since the exercise used a fictional recipient and controlled local interactions, this result should not be interpreted as evidence that users would fail to recognize or report phishing messages.

5.3 Overall Interpretation

The results indicate that the principal components of the local simulation operated as intended. GoPhish recorded the campaign activity, MailHog received the test email, the tracked campaign link opened the configured landing page, and the results dashboard displayed the corresponding interaction metrics.

The campaign recorded one email sent, one email opened, and one link clicked, with no submitted data or reported emails. The open and click events were generated through the author's controlled test activity. Consequently, the results demonstrate technical functionality within the laboratory environment rather than measurable employee or student susceptibility to phishing.

The zero data-submission count also aligns with the decision not to include a credential-collection form. This design reduced the risk of collecting sensitive information while allowing the core campaign workflow to be evaluated.

5.4 Limitations of the Results

The results should be interpreted within the following limitations:

1. **Small test scope:** The campaign used only one fictional recipient, so the results cannot be generalized to a larger population.
2. **Local email capture:** MailHog demonstrated local SMTP functionality but did not test external email delivery, spam filtering, or production email security controls.
3. **Controlled interactions:** The opening and click events came from the author's own test activity rather than an independent participant.
4. **Tracking limitations:** Open and click events show that tracking activity was registered, but they do not independently establish that a person read or understood the message.
5. **No credential collection:** The landing page did not request credentials, so the experiment did not evaluate credential-submission behavior.
6. **No reporting event:** The absence of a reported-email event provides no meaningful measure of real-world reporting behavior in this test.

5.5 Conclusion of Results

The local phishing-awareness simulation successfully demonstrated the basic integration of GoPhish, MailHog, campaign tracking, and a security-awareness landing page. The observed metrics were consistent with the intended test workflow, and no credential data was submitted.

The exercise provides evidence of successful technical configuration and controlled testing. A larger, authorized assessment with independent participants and appropriate safeguards would be required to draw conclusions about user awareness or phishing susceptibility.

Findings

The laboratory exercise produced several findings regarding the configuration, operation, and integration of GoPhish and MailHog. The observations below are based on the local tests performed and the results recorded during the campaign.

6.1 Successful GoPhish Initialization

GoPhish started successfully, with its administrative interface configured at https://127.0.0.1:3333 and its phishing server at http://127.0.0.1:8080.

This confirmed that the application could initialize its principal services and provide access to the campaign management interface. The local binding also limited the configured service endpoints to the local machine, reducing exposure to other devices on the network.

Significance: Successful initialization established the foundation required to create, launch, and monitor the controlled phishing-awareness campaign.

6.2 Successful MailHog Operation

MailHog accepted local SMTP connections through 127.0.0.1:1025 and provided a web interface at http://127.0.0.1:8025. Test messages appeared in its inbox and could be opened for inspection.

This demonstrated that MailHog was functioning as a local email-capture service and could be used to examine messages without relying on an external email provider.

Significance: MailHog provided a controlled environment for testing email delivery and inspecting message content while reducing the risk of unintentionally sending test emails to external recipients.

6.3 Successful Campaign Email Delivery to MailHog

The campaign email generated by GoPhish was captured in the MailHog inbox. The message could be inspected to review its subject, sender information, and content, including the campaign tracking link.

This finding demonstrated that the configured GoPhish sending profile could communicate with the local SMTP service and submit the campaign message for capture.

Significance: The result confirmed the basic integration between GoPhish and MailHog. It demonstrated local message delivery, not delivery to an external mailbox or successful passage through production email security controls.

6.4 Successful Landing Page Access

The campaign tracking URL opened the locally hosted Security Awareness Exercise page. The page displayed security-awareness guidance intended to encourage careful inspection of suspicious emails and links.

The page did not request credentials or provide a credential-submission form.

Significance: This confirmed that the campaign URL and configured landing page worked together as intended. It also demonstrated that a phishing-awareness exercise can provide educational content without collecting sensitive user information.

6.5 Successful Campaign Event Recording

The GoPhish dashboard recorded the following events for the single test recipient:

- **Emails sent:** 1
- **Emails opened:** 1
- **Links clicked:** 1

The opening and clicking events were generated during the author's controlled test activity. These events demonstrated that GoPhish registered the corresponding campaign interactions.

Significance: The results confirmed that the campaign's basic tracking functionality operated during the test. However, the events should not be interpreted as evidence of an independent participant's behavior or as a measure of real-world phishing susceptibility.

6.6 No Data Submission Recorded

The campaign dashboard recorded zero submitted-data events. This result was consistent with the awareness landing page's design, which did not request or collect credentials.

The exercise therefore demonstrated campaign tracking without requiring the collection of passwords or other sensitive information.

Significance: The result aligned with the project's safety objectives and reduced the risk of handling sensitive data during the simulation.

6.7 Overall Finding

The findings indicate that the local phishing-awareness laboratory was successfully configured for its intended testing purpose. GoPhish initialized correctly, MailHog captured the test email, the tracking URL opened the awareness page, and the campaign dashboard recorded the expected test interactions.

The exercise demonstrated the practical integration of email-simulation software, local SMTP testing, landing-page configuration, and campaign monitoring. Its results are limited to the controlled environment and single test recipient used. They establish technical functionality, not the effectiveness of phishing defenses across a wider population.

Overall, the laboratory exercise provided practical experience in configuring and operating phishing-simulation tools while maintaining a safety-focused approach that avoided credential collection.

Limitations

Although the laboratory exercise successfully demonstrated the basic operation of GoPhish and MailHog, several limitations affect the interpretation and generalizability of the results. These limitations should be considered when evaluating the outcomes of the project.

7.1 Limited Sample Size

The campaign used only one fictional test recipient, student1@example.test. Although this was sufficient to test the basic campaign workflow, a single test recipient does not represent a diverse group of users with different levels of cybersecurity awareness, experience, and behavior.

Consequently, the observed results cannot be generalized to students, employees, organizations, or other real-world populations. The exercise was intended to verify technical functionality rather than measure user susceptibility to phishing.

Impact: The findings demonstrate that the configured workflow operated during the test, but they do not establish a population-level awareness rate or phishing-click rate.

7.2 Limitations of Email-Open Tracking

Email-open tracking may rely on a tracking pixel or another email-client-dependent mechanism. Whether an open event is recorded can depend on image loading, privacy settings, automated image retrieval, and other client behaviors.

Therefore, an open event does not conclusively prove that a person read or understood the email. Similarly, the absence of an open event would not necessarily prove that the message was never viewed.

In this project, the recorded open event was associated with the author's controlled test activity.

Impact: Open-tracking results should be treated as indicators of recorded tracking activity, not definitive evidence of human attention or comprehension.

7.3 Local Testing Environment

The experiment was conducted using GoPhish and MailHog within a local laboratory environment. MailHog captured messages through its local SMTP service, allowing the campaign email to be inspected without relying on a public email provider.

The project did not evaluate delivery through external email services or test production email security mechanisms, such as enterprise spam filters, secure email gateways, domain authentication policies, or organizational mail-flow rules.

Impact: Successful delivery to MailHog does not guarantee that the same message would reach an external inbox or pass through real-world email security controls.

7.4 Absence of a Reported-Email Event

The campaign dashboard recorded zero reported emails. This indicates that no email-reporting event was recorded during the test.

However, the campaign used a fictional recipient and controlled interactions. No independent group of participants was assessed for its ability or willingness to identify and report suspicious messages.

Impact: The absence of a reporting event cannot be used to conclude that users would fail to report a real phishing email. Reporting behavior was not meaningfully evaluated by this experiment.

7.5 Limitations of Screenshot Evidence

Screenshots were collected to document the configuration screens, captured messages, landing page, and campaign results dashboard. These images provide visual evidence of the application state at the time they were captured.

However, screenshots are static records. They do not independently verify every underlying process, establish the complete sequence of events, or demonstrate that the same behavior would occur under different conditions. Their evidentiary value also depends on their clarity, context, and accurate labeling.

Impact: Screenshots should be considered supporting documentation alongside the described methodology and observed results, rather than conclusive proof of behavior beyond the laboratory environment.

7.6 Limited Evaluation of User Behavior

The exercise focused on configuring the tools and verifying email delivery, link tracking, and landing-page functionality. It did not measure participants' ability to recognize phishing indicators, their decision-making process, or their response to security-awareness guidance.

No conclusions can therefore be drawn about the effectiveness of a security-awareness program or the behavior of a broader user population.

Impact: The project demonstrates technical implementation and controlled testing rather than a comprehensive human-factors or security-awareness evaluation.

7.7 Overall Assessment of Limitations

The limitations do not negate the successful technical outcomes of the exercise. Instead, they define the boundaries of what can reasonably be concluded from the available evidence.

The project demonstrates that GoPhish and MailHog could be integrated in a local environment, that a test email could be captured, that the campaign URL could open the intended awareness page, and that GoPhish could record test interactions.

The results should remain limited to those observations. Evaluating real-world email delivery, user behavior, reporting rates, and organizational defenses would require a broader, explicitly authorized test design with appropriate safeguards and a larger sample.

Recommendations

Based on the results and limitations identified during the laboratory exercise, the following recommendations are proposed to improve the security, reliability, documentation, and educational value of future phishing-awareness simulations.

8.1 Maintain Localhost-Bound Services

GoPhish and MailHog should remain bound to localhost addresses when conducting isolated laboratory exercises. The GoPhish administrative interface, phishing server, and MailHog services should only be exposed to other devices when a separate, explicitly approved, and appropriately secured test environment requires it.

Administrative access should be protected, and default or weak credentials should be avoided. Unnecessary network exposure should be minimized, and services should be stopped when testing is complete.

Rationale: Restricting access reduces the likelihood of unauthorized interaction with the laboratory tools and helps prevent test emails or campaign pages from being exposed beyond their intended environment.

8.2 Use Fictional or Explicitly Authorized Test Recipients

Future exercises should continue to use fictional recipients or individuals who have been explicitly authorized to participate. Campaign scope, recipient lists, and testing objectives should be approved before a campaign is launched.

The exercise should not request, collect, or store real passwords or other sensitive account information. Where the purpose is to teach users how to recognize phishing, a dedicated awareness landing page should provide educational feedback without requiring credential submission.

Rationale: These practices reduce privacy and security risks while keeping the exercise focused on learning and controlled technical validation.

8.3 Improve Email Template Formatting

The captured campaign email showed HTML anchor markup displayed as literal text in the plain-text view. This indicates that the HTML content and plain-text content should be reviewed and configured separately.

For future campaigns, the HTML version should contain correctly formatted HTML elements, while the plain-text version should contain readable text and a visible URL where appropriate. The final message should be inspected in MailHog before launching a campaign.

Rationale: Correct formatting improves readability, prevents confusing message presentation, and helps ensure that the delivered test email matches the intended design.

8.4 Improve Evidence Management and Documentation

All screenshots and supporting evidence should be stored in a clearly organized project folder. Each item should have a unique evidence identifier, a descriptive filename, a capture date where available, and a caption explaining what it demonstrates.

Suggested evidence identifiers include:

- E01_GoPhish_Startup
- E02_MailHog_SMTP_Verification

- E03_Email_Template
- E04_Awareness_Landing_Page
- E05_Campaign_Overview
- E06_Campaign_Results

Before sharing screenshots, review them for passwords, authentication tokens, private addresses, personal information, or other sensitive details. Any such information should be redacted where necessary.

Rationale: Consistent evidence management improves report readability, makes findings easier to verify, and reduces the risk of unintentionally disclosing sensitive information.

8.5 Define Success Criteria Before Testing

Future laboratory exercises should define measurable success criteria before execution. For example, the criteria may include:

1. GoPhish starts without critical errors.
2. MailHog accepts the local SMTP connection.
3. A verification email appears in the MailHog inbox.
4. The campaign email is captured and displays correctly.
5. The campaign tracking URL opens the intended awareness page.
6. The campaign dashboard records the expected test events.
7. No credentials or unauthorized personal data are collected.

Each criterion should have a corresponding verification method and evidence reference.

Rationale: Predefined criteria make the test more systematic and provide a clear basis for determining whether the intended technical objectives were achieved.

8.6 Record Scope, Approval, Dates, and Cleanup Procedures

Before conducting a future exercise, document the approved scope, test objectives, authorized recipients, test dates, responsible individuals, and any restrictions imposed by the instructor or organization.

After testing, stop the relevant services when they are no longer needed, review and remove unnecessary test data, and retain only the evidence required for the project or assessment. Configuration files and logs should be reviewed before being shared or archived.

Rationale: Clear authorization and cleanup procedures support responsible testing, reduce unnecessary exposure, and improve the reproducibility of the laboratory exercise.

8.7 Strengthen Future Evaluation

If the project is expanded beyond a single-recipient technical test, the next stage should be designed to evaluate additional aspects of phishing awareness. Any exercise involving independent participants should have appropriate approval, clearly defined boundaries, and safeguards for participant privacy.

Evaluation criteria could include whether participants recognize suspicious indicators, access the educational landing page, or use the approved reporting process. Results should be aggregated where appropriate and interpreted within the limits of the test design.

Open and click events should not be treated as definitive evidence of human behavior without considering tracking limitations and the context in which the events occurred.

Rationale: A more structured evaluation could provide better evidence of learning outcomes while avoiding unsupported conclusions from limited tracking data.

8.8 Overall Recommendation

The existing laboratory setup is suitable for demonstrating the basic integration of GoPhish, MailHog, campaign tracking, and security-awareness content in a controlled environment. The immediate priorities are to maintain local-only service exposure, correct the email template formatting, preserve clear evidence, and document the test boundaries and cleanup process.

Future work should build on the current technical results through predefined success criteria and carefully scoped, authorized testing. These improvements would make the project more reproducible, professionally documented, and useful as a cybersecurity learning exercise.

Evidence Gallery

The following screenshots were provided during the lab workflow. Captions identify what each image demonstrates.

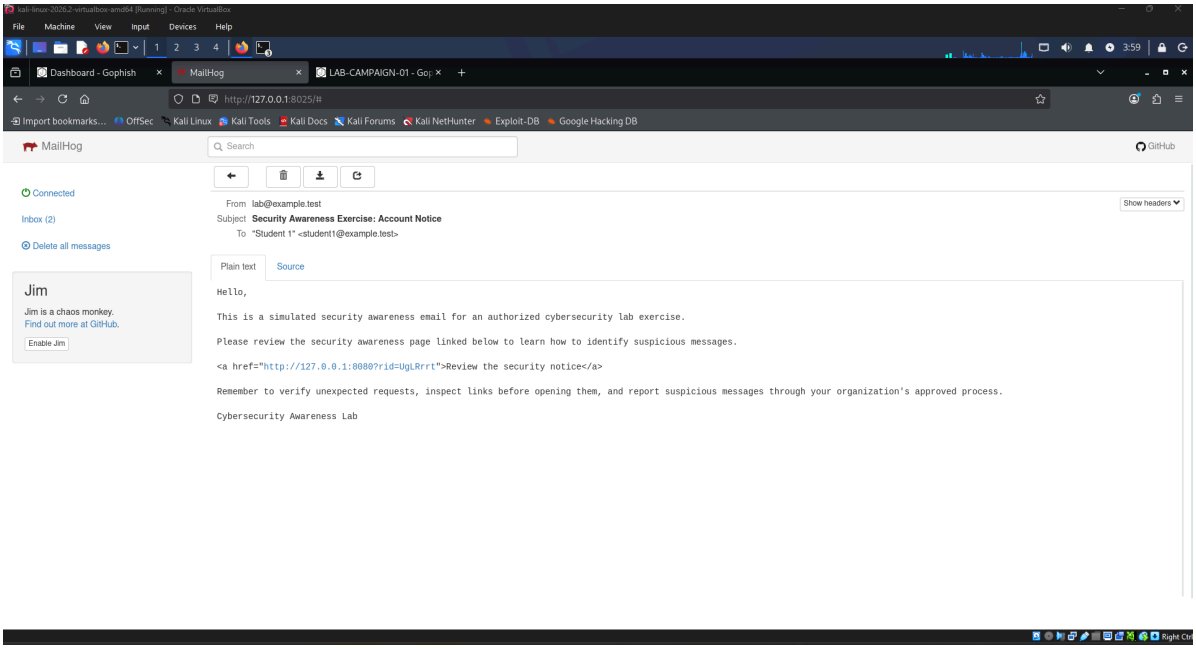


Figure 1: Campaign email captured in MailHog

MailHog shows the campaign email with the local tracking URL. The plain-text view displays the anchor markup literally, which supports the recommendation to maintain a separate plain-text version and HTML version.

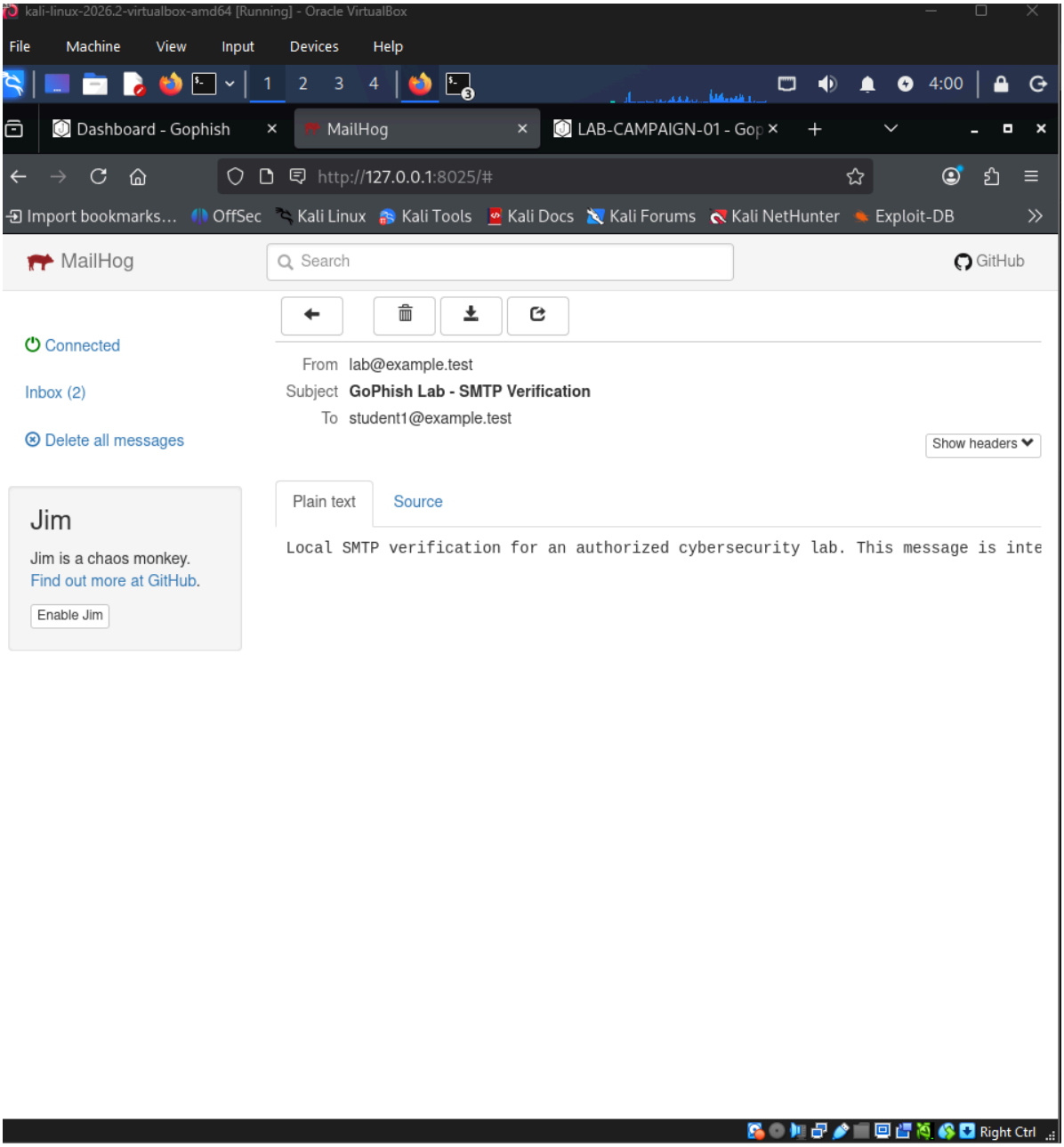


Figure 2: SMTP verification message

A direct SMTP test message appears in MailHog, confirming local SMTP delivery to the test inbox.

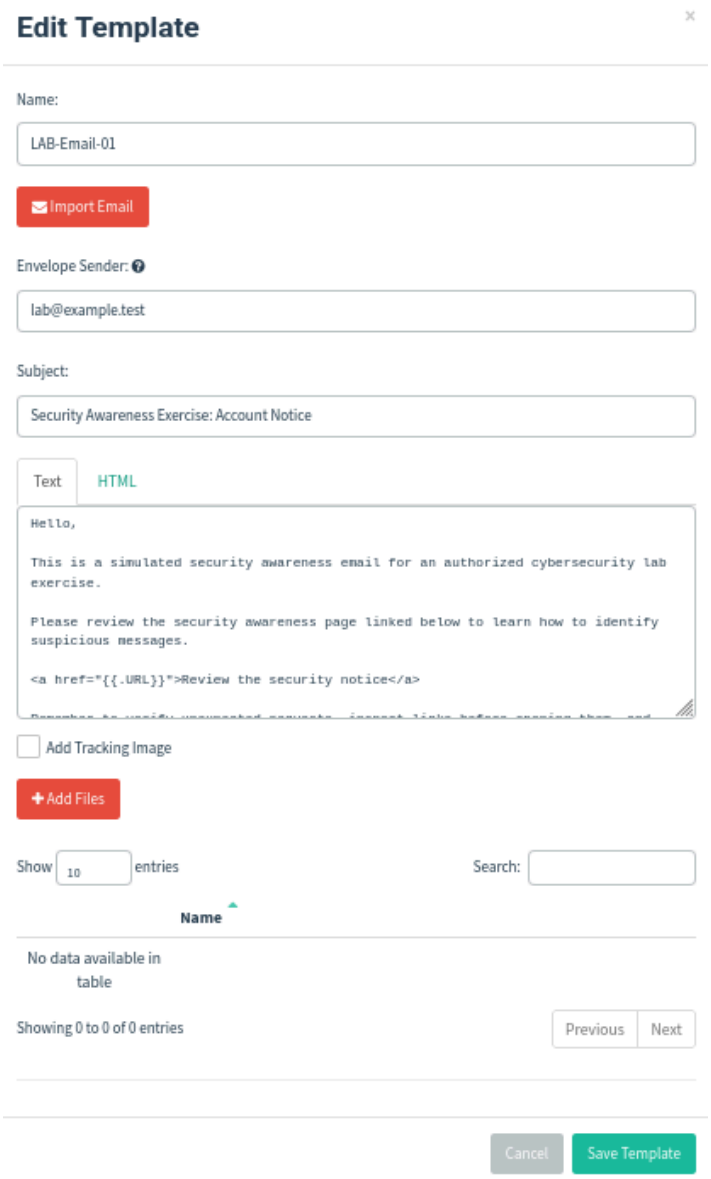


Figure 3: GoPhish email template editor

The LAB-Email-01 template configuration is visible. The Text tab is selected in this screenshot.

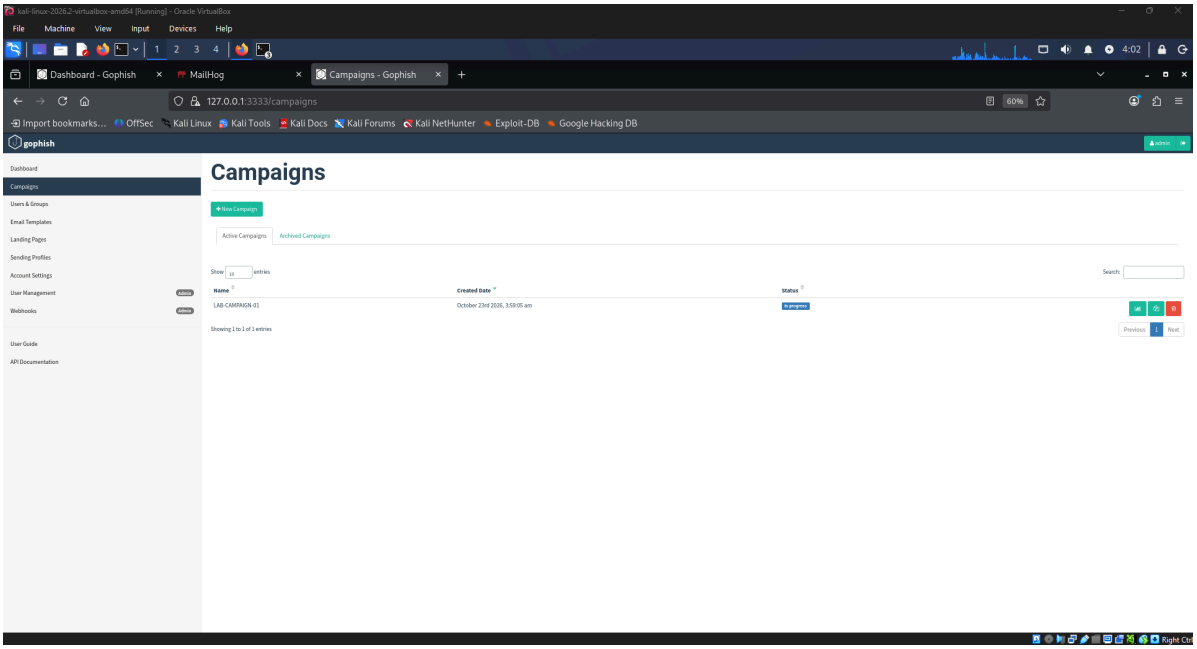


Figure 4: GoPhish campaign dashboard

The campaign list shows LAB-CAMPAIGN-01 with status In progress.

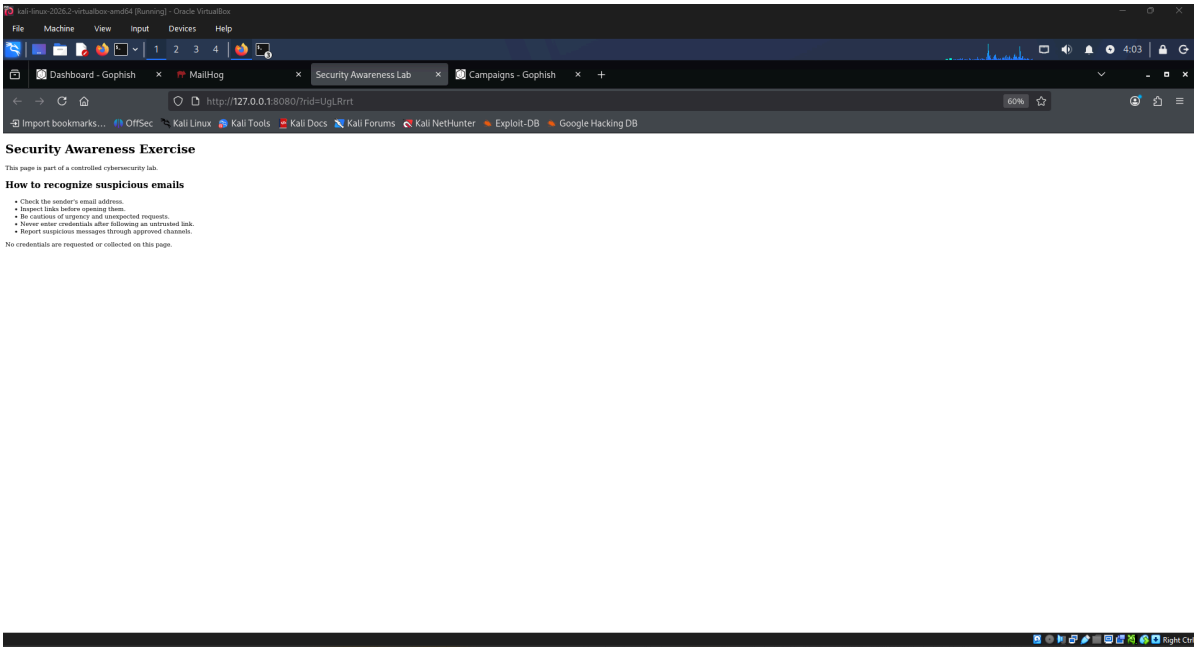


Figure 5: Security awareness landing page

The tracked localhost URL loads the awareness page, including advice on recognizing suspicious emails and a statement that no credentials are collected.

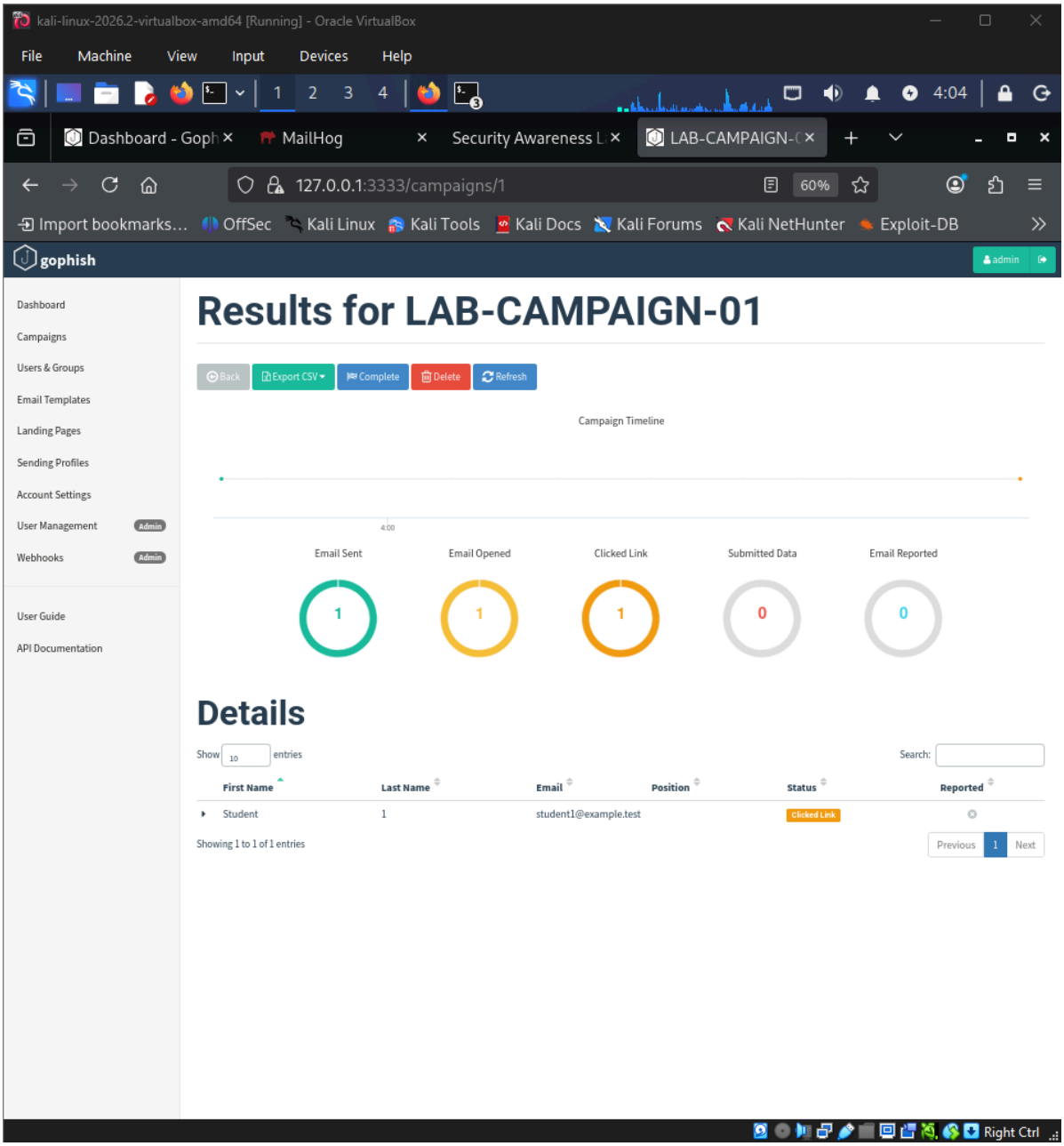


Figure 6: GoPhish campaign results

The dashboard records 1 email sent, 1 opened, 1 clicked, 0 submitted data, and 0 reported emails.

Conclusion

The Home Lab Project successfully demonstrated the basic configuration and operation of GoPhish and MailHog for a controlled security-awareness simulation in a local Kali Linux environment. The exercise provided practical experience in configuring a phishing-simulation platform, establishing a local email-testing service, creating an email template, configuring a security-awareness landing page, and monitoring campaign events.

The results confirmed that GoPhish started successfully, MailHog captured the test email, and the campaign tracking URL opened the intended Security Awareness Exercise page. The GoPhish dashboard recorded one email sent, one email opened, and one link clicked, with zero submitted-data events and zero reported emails. These observations demonstrated that the principal components of the local testing workflow operated as intended.

However, the results must be interpreted within the limitations of the experiment. The campaign involved only one fictional test recipient, and the opening and clicking events resulted from the author's own controlled test activity. Therefore, the recorded metrics represent technical validation of the laboratory setup rather than a real-world phishing success rate or a measurement of user susceptibility to phishing attacks.

An important aspect of the project was its emphasis on responsible cybersecurity testing. The services were configured for local use, a fictional recipient was used, and the awareness landing page provided educational guidance without requesting or collecting credentials. These measures helped maintain the exercise's focus on learning while reducing unnecessary security and privacy risks.

The project also highlighted areas for improvement, particularly email-template formatting, evidence organization, and the documentation of testing scope and success criteria. Addressing these areas would improve the clarity, reproducibility, and professionalism of future laboratory exercises.

Overall, this project strengthened practical understanding of email-based security-awareness simulations, local SMTP testing, campaign tracking, and the interpretation of cybersecurity test results. It established a foundation for further learning in phishing awareness, defensive security practices, and controlled security testing. Most importantly, the exercise demonstrated that technical findings must be evaluated carefully, documented accurately, and presented within the limits of the evidence collected.